

Computing and Interpreting Centre and Spread

Answer Key

Grade 7

Quick Answers

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|--|---|
| 1. 6 | 8. 16 |
| 2. 11 | 9. (a) interquartile range for Route A = 6, (b) interquartile range for Route B = 18, — |
| 3. 7 | 10. (a) the mean = 20, (b) the mean absolute deviation = 4.8 |
| 4. 3 | 11. 86 |
| 5. (a) the lower quartile = 6, (b) the upper quartile = 18, (c) the interquartile range = 12 | 12. (a) the mean = 10, (b) the median = 6, (c) the range = 42, (d) the interquartile range = 5, — |
| 6. (a) the mean = 20, (b) the median = 14, — | |
| 7. (a) the range = 22, (b) the interquartile range = 12 | |
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What is verified

9 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 6, 9, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 7

6.SP.A – *problems 6, 9, 12*

6.SP.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 24 tagged checks.

1. Mean of 4, 7, 5, 9, 5.

Solution: The five values total $4 + 7 + 5 + 9 + 5 = 30$, and $30 \div 5 = 6$.

6

2. Range of 11, 17, 9, 20, 14.

Solution: The largest value is 20 and the smallest is 9, so the range is $20 - 9 = 11$.

11

3. Median of 2, 9, 4, 11, 6, 8.

Solution: In order: 2, 4, 6, 8, 9, 11. There are six values, so the median is the mean of the third and fourth: $(6 + 8) \div 2 = 7$.

7

4. Mode of 3, 5, 3, 7, 3, 5, 8.

Solution: The value 3 appears three times, 5 appears twice, and the others once each, so the mode is 3. 3

5. Quartiles of 2, 4, 8, 10, 14, 16, 20, 22.

Solution: (a) Eight values split into a lower four and an upper four. Lower half 2, 4, 8, 10: median $(4 + 8) \div 2 = 6$. 6

(b) Upper half 14, 16, 20, 22: median $(16 + 20) \div 2 = 18$. 18

(c) Interquartile range $= 18 - 6 = 12$. 12

6. Bicycle prices 12, 14, 13, 15, 46 dollars.

Solution: (a) The prices total 100 dollars, and $100 \div 5 = 20$ dollars. 20

(b) In order: 12, 13, 14, 15, 46. With five values the median is the third, 14 dollars. 14

(c) **Instructor-judged.** The median. Four of the five bicycles are priced between 12 and 15 dollars, and the single 46 dollar bicycle drags the mean up to 20 — a price higher than four of the five actually on sale. Full credit needs the choice of the median plus the effect of the one unusually large price on the mean.

7. Range and interquartile range of 1, 3, 5, ..., 23.

Solution: (a) $23 - 1 = 22$. 22

(b) Twelve values split into a lower six and an upper six:

$$\underbrace{1, 3, 5, 7, 9, 11}_{\text{lower quartile 6}} \quad \underbrace{13, 15, 17, 19, 21, 23}_{\text{upper quartile 18}}$$

The interquartile range is $18 - 6 = 12$, far smaller than the range because it ignores both ends. 12

8. Five numbers have mean 12; four of them are 8, 10, 15, 11.

Solution: A mean of 12 over five numbers means the total is $12 \times 5 = 60$. The four known numbers total $8 + 10 + 15 + 11 = 44$, so the fifth is $60 - 44 = 16$. Written as an equation, $\frac{8 + 10 + 15 + 11 + x}{5} = 12$. x = 16

9. Two bus routes, both with median 14 minutes.

Solution: (a) Route A in order is 10, 12, 14, 16, 18. Leaving out the middle value, the lower half is 10, 12 and the upper half is 16, 18, so the quartiles are 11 and 17 and the interquartile range is $17 - 11 = 6$ minutes. 6

(b) Route B is 2, 8, 14, 20, 26. The quartiles are $(2 + 8) \div 2 = 5$ and $(20 + 26) \div 2 = 23$, so the interquartile range is $23 - 5 = 18$ minutes. 18

(c) **Instructor-judged.** Route A is more predictable. Both routes have the same median, so the centre cannot separate them; the spread can. The middle half of Route A's times covers only 6 minutes while Route B's covers 18. Full credit needs the comparison of the two interquartile ranges, not just the observation that the medians match.

10. Delivery masses 12, 16, 20, 24, 28 kilograms.

Solution: (a) The masses total 100 kilograms, and $100 \div 5 = 20$ kilograms. 20

(b) The distances from the mean are 8, 4, 0, 4, 8 kilograms. Their total is 24, and $24 \div 5 = 4.8$ kilograms. That is the typical distance a delivery sits from the mean. 4.8

11. Groups of 4 at mean 80 and 6 at mean 90; find the combined mean.

Solution: Means cannot simply be averaged when the groups differ in size. Recover the totals: $4 \times 80 = 320$ and $6 \times 90 = 540$, so all ten students total 860 marks and the combined mean is $860 \div 10 = 86$. Averaging 80 and 90 to get 85 would over-weight the smaller group. 86

12. Nine donations 2, 4, 4, 6, 6, 6, 8, 10, 44 dollars.

Solution: (a) The donations total 90 dollars, and $90 \div 9 = 10$ dollars. 10

(b) The values are already in order and there are nine, so the median is the fifth: 6 dollars. 6

(c) Range = $44 - 2 = 42$ dollars. 42

(d) Leaving out the median, the lower half is 2, 4, 4, 6 with median 4, and the upper half is 6, 8, 10, 44 with median $(8 + 10) \div 2 = 9$. The interquartile range is $9 - 4 = 5$ dollars. 5

(e) **Instructor-judged.** The median with the interquartile range. The single 44 dollar donation lifts the mean to 10 dollars — more than seven of the nine donations — and stretches the range to 42, while the median stays at 6 and the interquartile range at 5, which is what the bulk of the donations actually look like. Full credit needs the chosen pair together with an explanation naming the effect of the one large donation on both the mean and the range.
